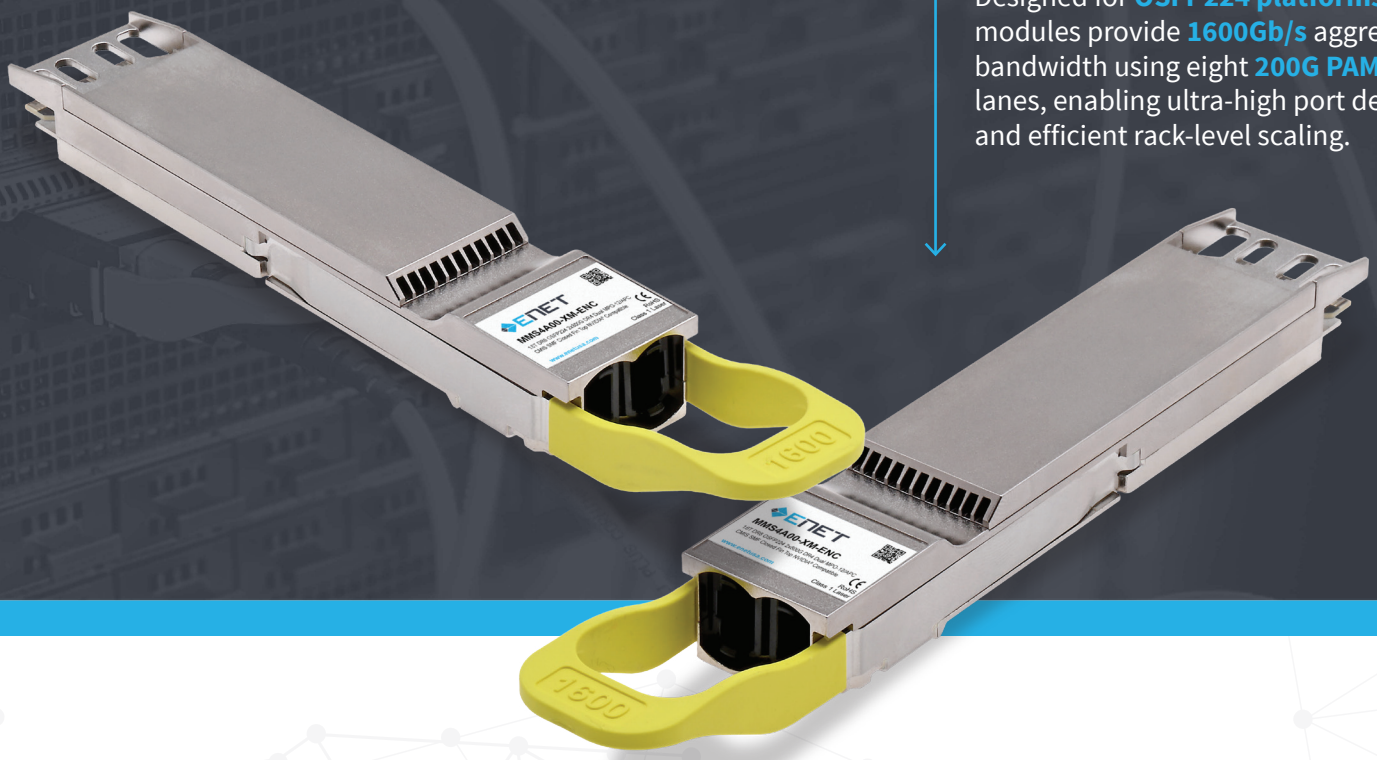
1.6T OEM COMPATIBLE TRANSCEIVERS

Built for AI, HPC, and Next-Generation Data Center Fabrics

As AI clusters grow and network speeds move beyond **800G,** infrastructure must scale without compromise. **1.6T optical transceivers** deliver the density, throughput, and signal integrity required for high-performance compute fabrics and hyperscale environments.

Designed for **OSFP224 platforms,** 1.6T modules provide **1600Gb/s** aggregate bandwidth using eight **200G PAM4** lanes, enabling ultra-high port density and efficient rack-level scaling.



Deployment Advantages:

- **High Density** - Maximize throughput per port with 1.6T aggregate bandwidth
- **Scalable Architecture** - Support next-generation switching platforms and high-speed aggregation layers
- **Optimized for Single-Mode Fiber** - Up to 500 meters over SMF at 1310nm
- **Operational Flexibility** - Dual optical interfaces support advanced fabric design strategies

Ideal Applications:

- Hyperscale data centers
- AI training clusters
- Machine learning infrastructure
- HPC compute environments
- High-speed core switching

1.6T transceivers are engineered for environments where bandwidth, stability, and density drive performance outcomes.

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Performance Overview:

- 1.6Tb/s aggregate throughput
- 8 × 200G PAM4 electrical lanes
- 8 × 200G optical lanes
- 1310nm single-mode operation
- Up to 500m reach over SMF
- Dual MPO-12/APC optical interface
- OSFP hot-pluggable form factor
- CMIS compliant management interface
- Operating case temperature: 0°C to 70°C

Each lane operates at 106.25 GBd, delivering the bandwidth required for modern AI and HPC interconnects.

Designed for High-Performance Architectures:

- AI cluster interconnects
- GPU server fabrics
- InfiniBand and Ethernet environments
- Spine-leaf architectures
- High-density switch-to-switch links
- 800G breakout architectures

Integrating dual 800G DR4 optical ports within a single OSFP224 module, these transceivers provide architectural flexibility while maintaining signal integrity across 500-meter single-mode links.

Signal Integrity and Link Reliability:

- Pre-FEC BER performance of 1×10^{-6}
- Post-FEC BER performance of 1×10^{-12} (host dependent)
- PAM4 modulation
- Per-lane transmit and receive optical monitoring
- Bias current monitoring
- Voltage and temperature monitoring
- FEC provided by the host system enables maximum supported reach

Advanced Diagnostics and Control:

- With CMIS-compliant management, host systems can monitor and control:
- Tx and Rx optical power per lane
- Bias current per lane
- Case temperature
- Supply voltage
- Loopback functions
- PRBS generator and checker capabilities
- Firmware upgrade functionality

This level of visibility is essential for AI and HPC deployments where uptime and rapid troubleshooting are critical.